

VIROLOGY

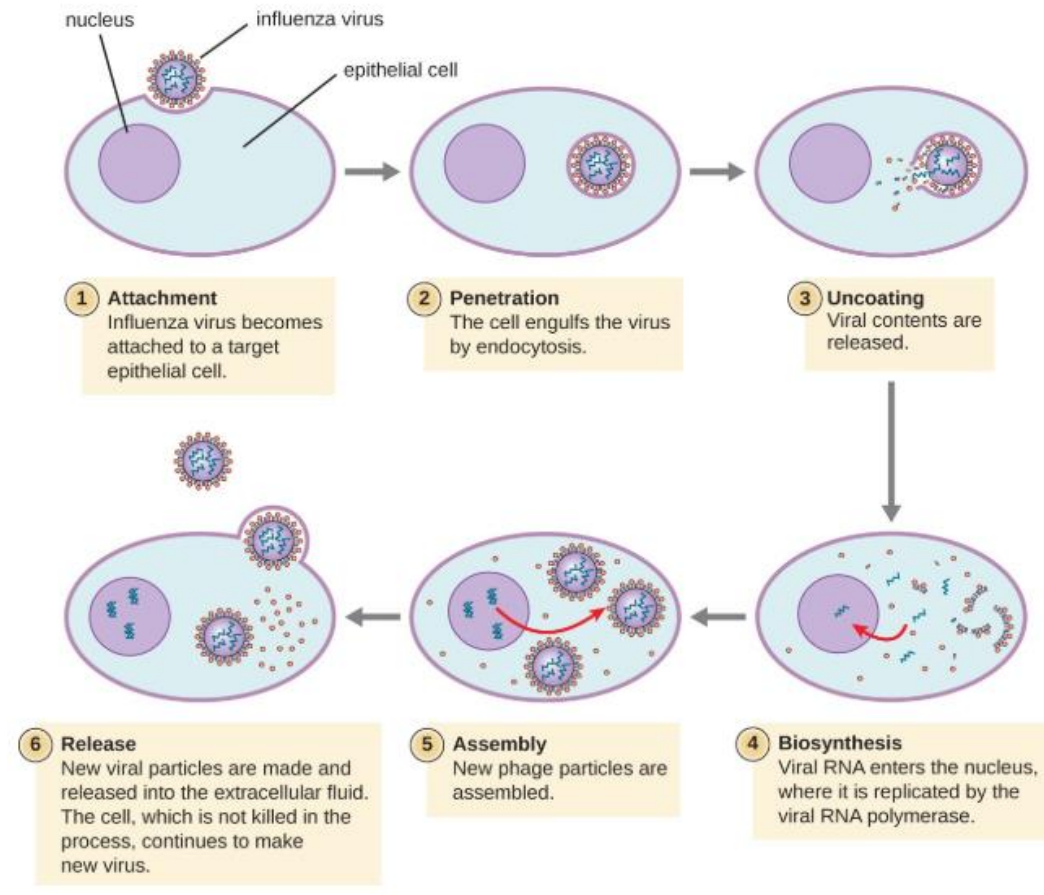
BY

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General Characteristics of Viruses

- **Viruses are obligate intracellular parasites** – cannot replicate on their own and must enter a host cell to replicate.
- Once inside living cells, viruses induce the host cell to **synthesize virus particles**.
- The **genome** of viruses can be either **DNA or RNA** (single or double-stranded).
- **Viruses do not have a system to produce ATP** – they rely on the host cell's energy production.
- **Size range of viruses:** Viruses can vary from **25 to 270 nm** in size.
- **Host specificity/ Tropism**
- The **classification of viruses** is based on the following:
 - **Nucleic acid type** (DNA or RNA)
 - **Size and shape of virion**
 - **Presence or absence of an envelope**

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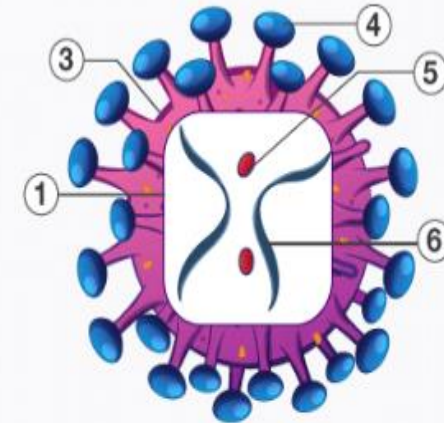
DNA and RNA Viruses

DNA Viruses	RNA Viruses
They are viruses that possess deoxyribonucleic acid (DNA) as their genetic material.	They are viruses that possess ribonucleic acid (RNA) as their genetic material.
The two types include – dsDNA and ssDNA viruses.	The two types include – dsRNA and ssRNA viruses.
They have a larger genome size.	They have a comparatively smaller genome size.
They show a low mutation rate.	They show a higher mutation rate.
Most DNA viruses are assembled in the nucleus.	Most RNA viruses are assembled in the cytoplasm.
They are included in the groups I, II, and VII of the Baltimore classification.	They comprise groups III, IV, V, and VI of the Baltimore classification.
Examples – Poxviruses, Herpesviruses and Papillomaviruses.	Examples – Retroviruses, Reoviruses, Picornaviruses etc.

VIRAL STRUCTURE



(a) Bacteriophage



(b) Human Immunodeficiency Virus

- 1 Capsid |
 2 DNA |
 3 Envelop |
 4 Envelop Protein |
 5 Enzyme |
 6 RNA

Baltimore Classification System

- The **Baltimore Classification System** classifies viruses based on their **nucleic acid genome** and **replication strategy**.
- It includes viruses that have different forms of RNA, such as **positive-sense** and **negative-sense**, and those that can **reverse transcribe** RNA into DNA or vice versa.
- Developed by **David Baltimore** in the 1970s.
- **Categorizes viruses** based on:
 - Type of **nucleic acid genome**
 - **Replication strategy** of the virus
- The system includes **7 classes**, with special focus on RNA viruses and their replication mechanisms.

Key Concepts in the Baltimore System

- **Positive-strand RNA (+RNA):** This RNA can be immediately translated into proteins (e.g., mRNA in cells).
- **Negative-strand RNA (–RNA):** This RNA is not directly translatable into proteins; it must first be transcribed into positive-strand RNA.
- **Reverse Transcription:** Some viruses can **reverse transcribe**, meaning they can create **DNA from an RNA template**, a process not naturally occurring in host cells.

The 7 Classes of Viruses

- **Class I: dsDNA Viruses**

- Double-stranded DNA viruses.
- Example: Herpesvirus

- **Class II: ssDNA Viruses**

- Single-stranded DNA viruses.
- Example: Parvovirus

- **Class III: dsRNA Viruses**

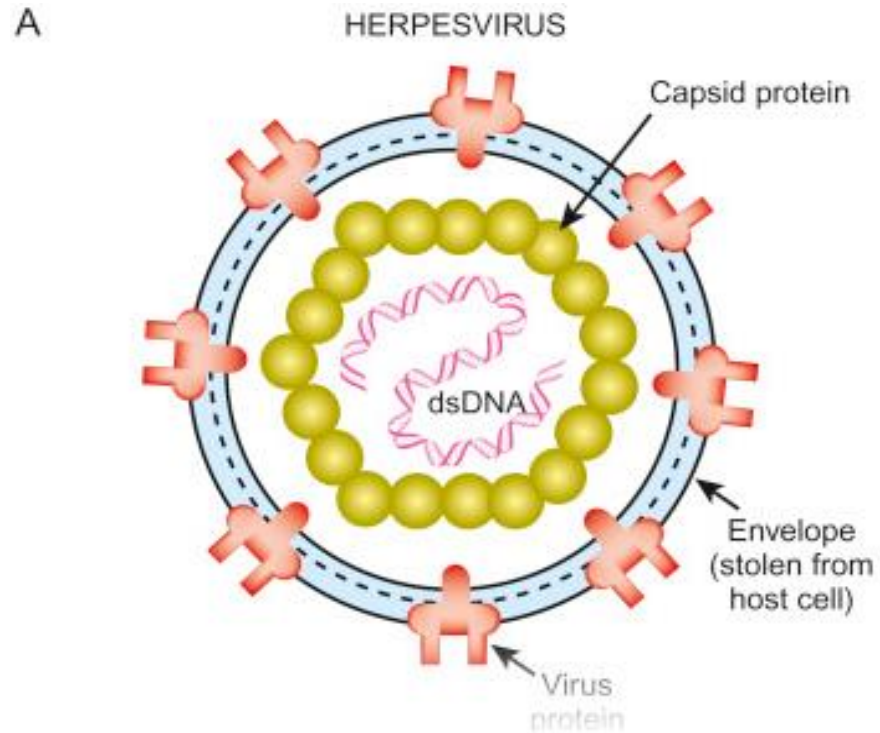
- Double-stranded RNA viruses.
- Example: Rotavirus

- **Class IV: Positive-sense ssRNA Viruses**

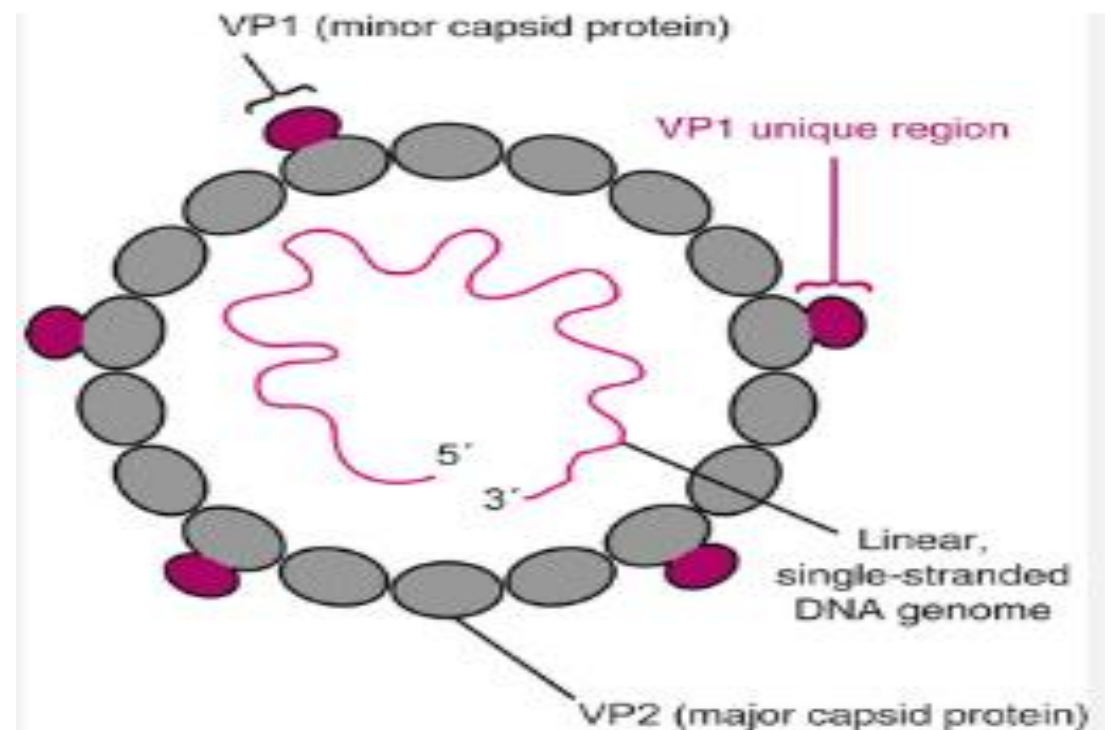
- Single-stranded RNA viruses that are positive-sense and can be directly translated into proteins.
- Example: Picornavirus

DNA Viruses

Double Stranded DNA Virus



Single Stranded DNA Virus



Classes of Viruses Conti...

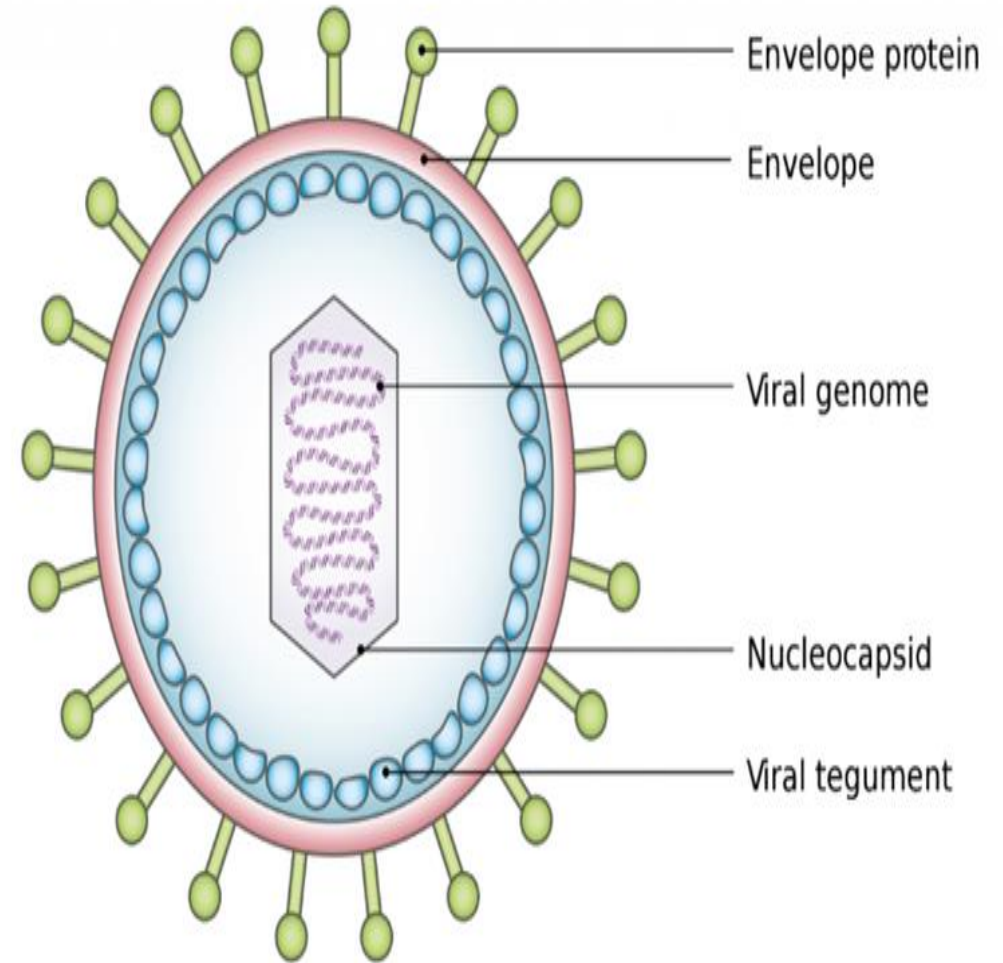
- **Class V: Negative-sense ssRNA Viruses**
 - Single-stranded RNA viruses that are negative-sense, requiring transcription into positive-sense RNA before translation.
 - Example: Influenza virus
- *Suggested Diagram:* Diagram showing the transcription process from negative-sense to positive-sense RNA.
- **Class VI: RNA Viruses That Reverse Transcribe**
 - RNA viruses that can reverse transcribe, converting RNA into DNA.
 - Example: Retrovirus (e.g., HIV)
- **Class VII: DNA Viruses That Reverse Transcribe**
 - DNA viruses that use reverse transcription to convert RNA back into DNA.
 - Example: Hepatitis B virus

Viral Structure

- **Virion** is the entire viral particle, the complete virus that exists outside the host cell.
- **Capsid**: The **protein coat** that encloses the genetic material of the virus.
- **Capsomer**: The **protein subunit** that makes up the capsid. These are arranged in a repeating pattern to form the capsid structure.
- **Nucleocapsid**: A complex formed by the **capsid and genetic material** (nucleic acid).
- **Envelope**:
 - The outer coating of some viruses made of a **phospholipid bilayer**.
 - The envelope contains **viral-encoded glycoproteins** and, in some cases, **viral-encoded matrix proteins**.
 - The envelope is derived from the host cell's membrane (it could come from the plasma membrane, endoplasmic reticulum, Golgi, or nuclear membranes).
 - **Naked nucleocapsids** are viruses that **do not have an envelope**.

Viral Structure

- **Essential Components:**
 - **Genome** (DNA or RNA)
 - **Capsid** (Protein coat)
 - **Envelope** (Optional: Lipid layer)
- **Genome:** DNA or RNA, single-stranded or double-stranded
- **Capsid:** Made of protein subunits (capsomers); shapes: helical, icosahedral, complex
- **Envelope:** Lipid bilayer derived from host cell membrane (not present in all viruses)



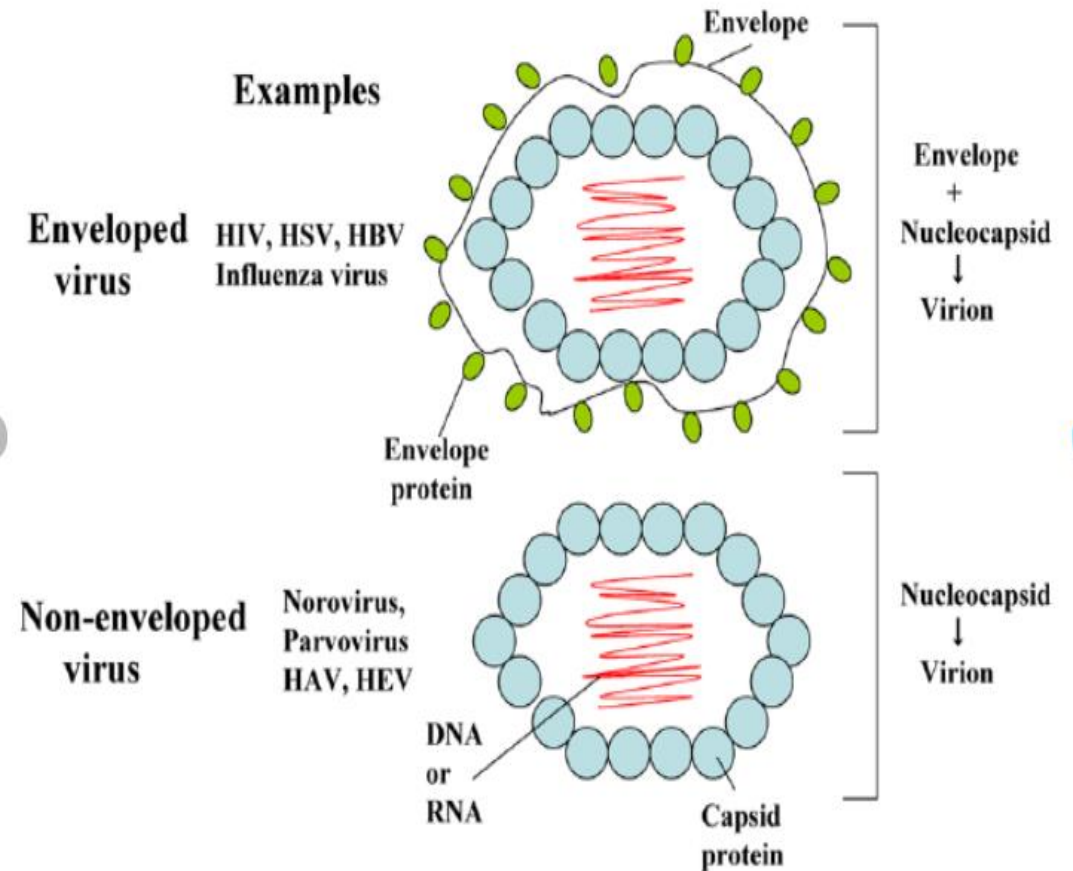
Types of Viruses Based on Structure

- **Naked (Non-enveloped) Viruses:**

- Capsid and genetic material only
- Examples: Poliovirus, Adenovirus

- **Enveloped Viruses:**

- Lipid membrane with glycoproteins
- Examples: Influenza, HIV, Herpesviruses

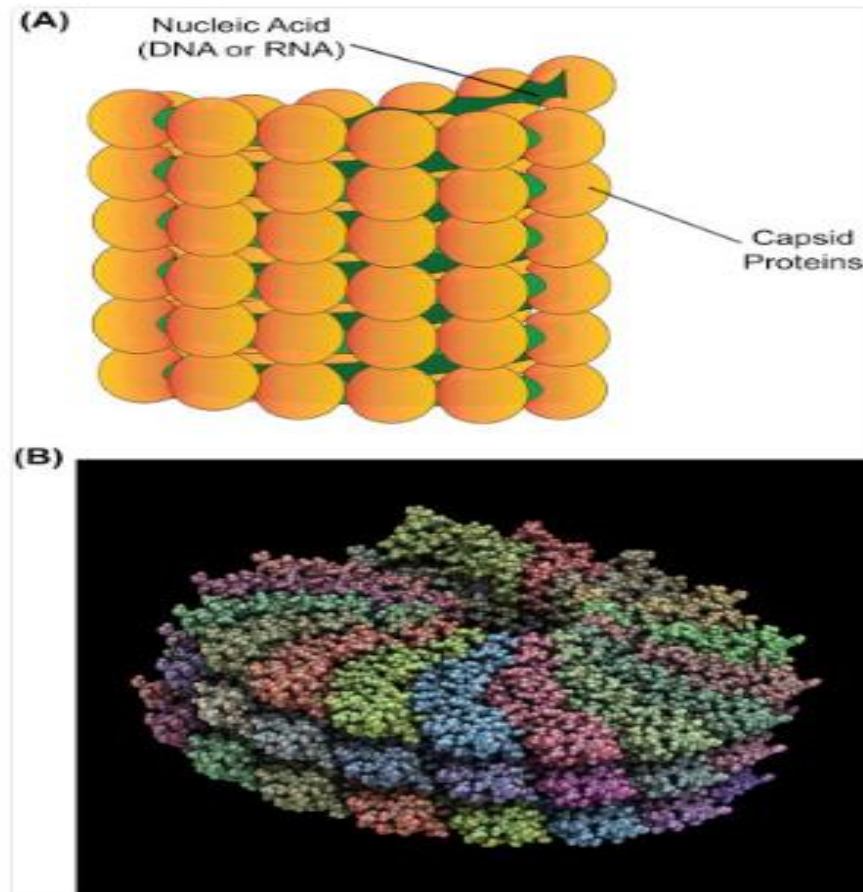


The types capsid and Function

- **Capsid:** A protein coat that surrounds and protects the viral **nucleic acid genome** from environmental damage.
- The capsid ensures that the virus can maintain its structure and integrity until it reaches a suitable host cell.
- **Types of Capsid Shapes**
- Virus capsids predominantly come in two main shapes:
 - **Helical** (spiral-shaped)
 - **Icosahedral** (symmetrical, 20-sided shape)

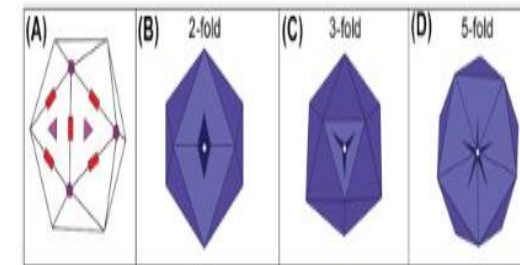
Common Types of Viral Capsids

- Helical capsid



- Icosahedral

Figure 2.6.



Icosahedron terminology and axes of symmetry.

(A) Icosahedron faces (fuchsia triangles), edges (red rectangles), and vertices (violet pentagons) are indicated on the white icosahedron. (B) The twofold axis of symmetry occurs when the axis is placed through the center of an edge. The threefold axis occurs when the axis is placed in the center of a face (C), and the fivefold axis passes through a vertex of the icosahedron (D).

Helical Capsid Structure

-
- The **helix** is a spiral shape that curves around an axis.
- Common biological structures also feature helices, such as:
 - Many **proteins** (e.g., alpha helices in protein structures)
 - The **DNA double helix**
- In **helical viruses**, the **nucleic acid** coils into a helical shape.
- **Capsid proteins** wind around the nucleic acid, forming a long **tube or rod-like structure** (called a nucleocapsid)
- The protein surrounding the nucleic acid is often called the **nucleocapsid protein**.
- Once the virus enters a host cell, the **helical nucleocapsid uncoils**, and the viral **nucleic acid becomes accessible** to the host's machinery.

Icosahedral Capsid Structure

- The **icosahedron** is a **symmetrical** shape with **20 triangular faces**, 12 vertices, and 30 edges.
- This shape provides the capsid with **maximum stability** and is the most efficient geometric shape for enclosing a viral genome.
- **Icosahedral viruses** typically have a **rigid structure**, and the capsid proteins are arranged in a precise, repeating pattern across the faces of the icosahedron.
- **Example:** Adenovirus and Poliovirus are examples of viruses with **icosahedral capsids**.

Virus Replication Process

1. Adsorption

- Attachment of the virus to a specific receptor on the host cell.

2. Penetration

- Entry of the virus into the host cell.

3. Uncoating

- Separation of the capsid from the genome or rearrangement of capsid proteins, exposing the genome for transcription and replication.

4. Eclipse Period

- Stage where genetic material is replicated, but intact virions are not yet detectable.
 - a. **Viral DNA/RNA** serves as the template for mRNA production.
 - b. **mRNA** codes for viral proteins and enzymes necessary for nucleic acid synthesis.

5. Assembly (Maturation)

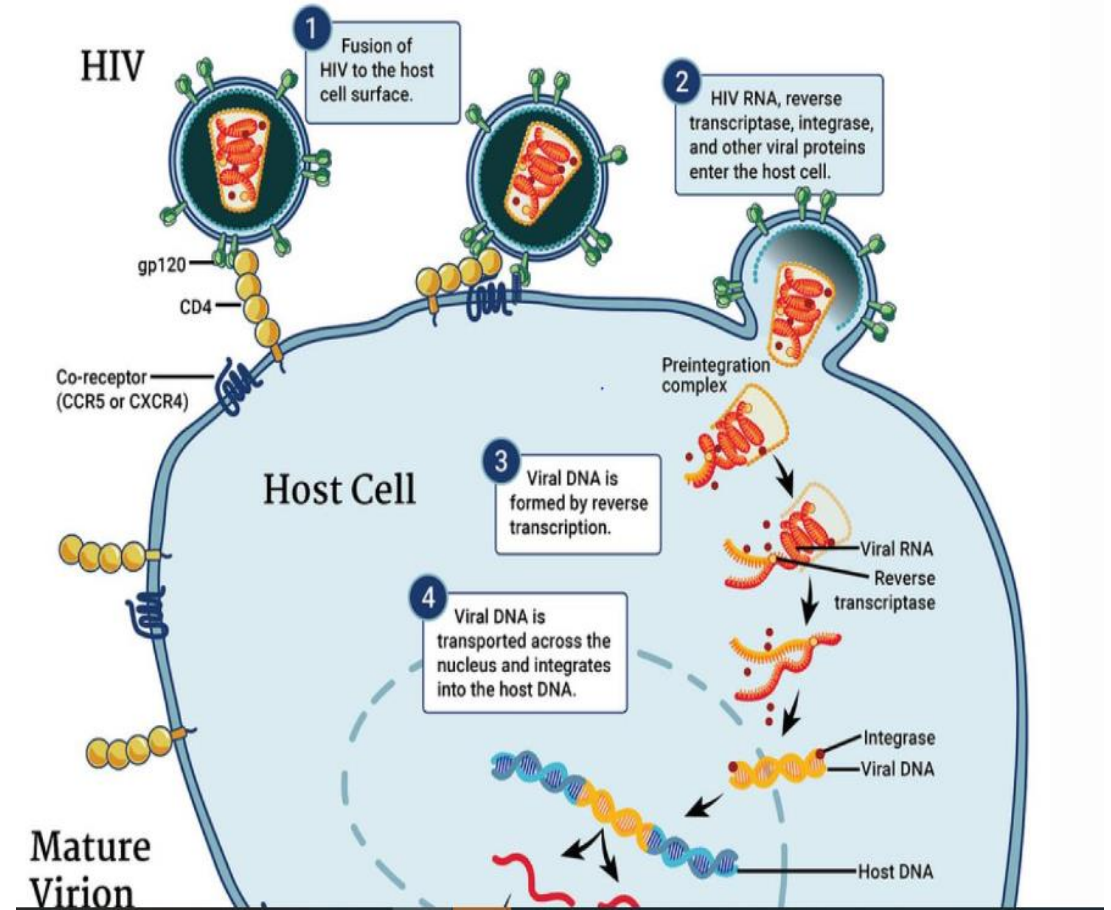
- Genetic material is assembled into a protein coat.

6. Release of Virions

- **Cell Lysis:** Naked viruses cause host cell lysis, leaving through a hole in the plasma membrane.
- **Budding:** Intact virions push outward from the host membrane. The membrane wraps around the virion, which is cleaved and resealed to form the viral envelope.

Viral Replication Cycle conti...

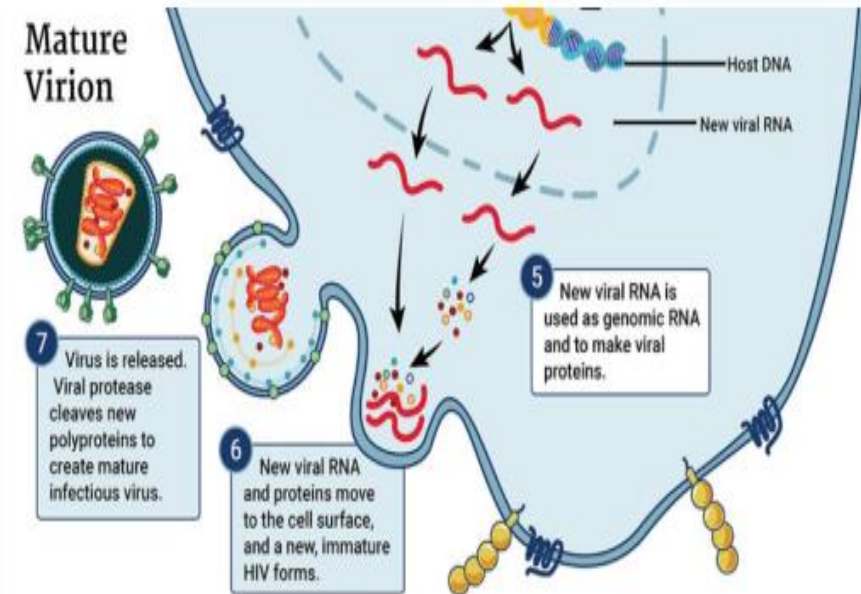
- **General Properties of Viruses**
- **Size:** 20-300 nm, only visible under electron microscopy
- **Classification Criteria:**
 - Type of genetic material (DNA or RNA)
 - Strand nature (single or double)
 - Capsid shape
 - Presence of envelope
- **Replication Cycle:** Attachment → Entry → Uncoating → Replication → Assembly → Release



Viral Replication Cycle conti...

- **Step 1:** Attachment - Virus binds to host cell receptors
- **Step 2:** Entry - Fusion or endocytosis
- **Step 3:** Uncoating - Capsid disassembled to release genome
- **Step 4:** Replication/Transcription - Host machinery used to replicate viral genome
- **Step 5:** Assembly - New virions are assembled
- **Step 6:** Release - Lysis (non-enveloped) or budding (enveloped)

HIV replication cycle



Specimen Collection and Processing for Diagnosing Viral Diseases

- **Viruses are most concentrated early in the disease course**
 - Samples should be collected **early** in the disease (first few days after symptom onset)
 - Viruses are in **highest concentrations** during this time

Sample Collection by Infection Site

- **Skin Infections:**
 - Rash site, **serum**, and **urine**
- **Respiratory Infections:**
 - **Sputum** or **throat swabs**
- **Central Nervous System (Meningitis & Encephalitis):**
 - **Cerebrospinal Fluid (CSF)**, **serum**, **stool**, and **throat swabs**
 - Brain biopsy (for encephalitis)
- **Urogenital Infections:**
 - **Needle aspirates**, **endocervical**, and **urethral swabs**
- **Gastrointestinal Tract:**
 - **Stool** samples and **rectal swabs**
- **Eye Infections:**
 - **Eye swabs** and **corneal scrapings**

Sample Transport for Viral Culture

- **Viral Transport Medium (VTM)**
 - Essential for transporting viral specimens for culture

Components of Viral Transport Medium (VTM)

- **Buffered saline**
- **Protein stabilizers**
- **Antimicrobials** to prevent bacterial and fungal growth
- **Storage and Handling of Viral Samples**
 - **Viral culture samples** should be refrigerated in VTM for **48 hours**
 - **Avoid freezing at -20°C**
 - **Storage at -70°C** is possible, but may **diminish infectivity**

Viral infection Sample sites and Associated Viral Agent

A. Respiratory System

- **Upper respiratory tract infections**

- **Viruses:** Rhinovirus, influenza virus, parainfluenza virus, respiratory syncytial virus (RSV), Epstein-Barr virus (EBV), coronavirus.
- **Clinical Diagnosis:** Nasopharyngeal swabs, polymerase chain reaction (PCR) tests, antigen detection, or viral cultures.

- **Croup and bronchitis**

- **Viruses:** Influenza virus, parainfluenza virus, RSV, adenovirus.
- **Clinical Diagnosis:** Clinical assessment, PCR, viral culture, or rapid antigen tests from throat or nasopharyngeal swabs.

- **Pneumonia in children**

- **Viruses:** RSV, parainfluenza virus, adenovirus, varicella-zoster virus (VZV).
- **Clinical Diagnosis:** Chest X-ray, PCR of respiratory samples (e.g., nasopharyngeal swab), or viral culture.

- **Pneumonia in adults**

- **Viruses:** Influenza virus, VZV, cytomegalovirus (CMV), RSV.
- **Clinical Diagnosis:** Chest X-ray, sputum culture, PCR, or serology.

B. Viral Meningitis

- **Viruses:** Enterovirus, echovirus, herpes simplex virus type 1 (HSV-1), HSV-2, VZV.
- **Clinical Diagnosis:** Lumbar puncture (cerebrospinal fluid [CSF] analysis), PCR, or viral culture of CSF.
- Elevated white blood cells, normal glucose levels, and mildly elevated protein in CSF suggest viral meningitis.

C. Encephalitis

- **Viruses:** HSV, VZV, and arboviruses (Togaviridae: eastern and western equine viruses, Flaviviridae: St. Louis encephalitis virus), rabies virus.
- **Clinical Diagnosis:** Magnetic resonance imaging (MRI)/computed tomography (CT) scan of the brain, lumbar puncture with CSF PCR, and viral culture from CSF.
- Brain biopsy may be performed in severe cases.

D. Cutaneous Infections

- **Viruses:** HSV-1, HSV-2, VZV, echovirus, measles virus, rubella virus, enterovirus, molluscum contagiosum virus, parvovirus B-19.
- **Clinical Diagnosis:** Skin swab or biopsy for viral PCR, direct fluorescent antibody testing, or viral culture.
- Clinical appearance (rash, vesicles) often aids diagnosis.

E. Genital Infections

- **Viruses:** HSV-2, human papillomavirus (HPV).
- **Clinical Diagnosis:** PCR or viral culture from genital lesions or swabs, serologic tests for HSV-2, and HPV DNA testing (for cervical screening).

F. Gastroenteritis

- **Viruses:** Rotaviruses, Norwalk viruses, adenoviruses, caliciviruses.
- **Clinical Diagnosis:** Stool samples for PCR, antigen detection, or viral culture.
- Enzyme-linked immunosorbent assay (ELISA) or immunoassays for rotavirus or Norwalk virus are commonly used.

G. Eye Infections

- **Viruses:** HSV, adenovirus, VZV.
- **Clinical Diagnosis:** Conjunctival or corneal scrapings, PCR, or viral culture. Fluorescein staining of cornea for suspected HSV keratitis.

H. Neonatal Infections

- **Viruses:** HSV, CMV, rubella virus.
- **Clinical Diagnosis:** Blood, CSF, or urine PCR for HSV, CMV, or rubella; viral culture; serologic testing for CMV (IgM/IgG) or rubella.
- Newborns may also undergo hearing screening for congenital CMV.

Cytological and histological diagnosis of Viral infection

DNA vs. RNA Virus Replication Sites

- **DNA Viruses:**
 - Most replicate in the **nucleus**, often producing **nuclear inclusions**.
 - Some DNA viruses assemble **elsewhere** in the cell.
- **RNA Viruses:**
 - Replicate and assemble in the **cytoplasm**, leading to **cytoplasmic inclusions**.
- See the table below

Viral Inclusions in Cells

Virus	Type of Inclusions	Location of Inclusions	Diagnostic Feature
HSV	Intranuclear inclusions	Nucleus	Nuclear inclusions indicative of herpes simplex virus
VZV	Intranuclear inclusions	Nucleus	Nuclear inclusions indicative of varicella-zoster virus
CMV	Enlarged cells (cytomegalic) with "owl eye" inclusions	Nucleus	Basophilic intranuclear inclusion resembling "owl eyes"
Other DNA Viruses	Nuclear inclusions	Nucleus	Typically caused by DNA viruses replicating in the nucleus
RNA Viruses	Cytoplasmic inclusions	Cytoplasm	RNA viruses replicate and assemble in the cytoplasm



Viral Isolation

- **What is Viral Isolation?**

- The process of growing viruses in a laboratory setting to help diagnose infections.
- It's an important step in understanding how viruses behave and how to treat or prevent them.

Cell Culture

- **What is Cell Culture?**

- A method of growing viruses in cells taken from animals, humans, or other organisms.
- Cells need to be in a special environment with nutrients, like serum from cows, to survive and grow.
- Different viruses prefer to infect certain types of cells (like lung cells, nerve cells, etc.), so it's important to use the right type of cells for the virus being studied.

Primary Cell Culture

- **What are Primary Cell Cultures?**

- Primary cell cultures are cells that are directly isolated from the tissues of an organism, such as animals or humans.
- These cells are **diploid**, meaning they have the normal number of chromosomes, similar to those in the original tissue.

- **How Are They Made?**

- Cells are taken from a living organism's tissue (e.g., kidney, liver, or lungs) and placed in a culture medium that provides essential nutrients for growth.
- The cells will grow for a short period before they stop dividing and eventually die.

- **Why Are They Important?**

- Primary cultures are often the best choice for growing viruses that require a more natural environment.
- They are **permissive** to many viruses, meaning they support viral replication.
- **Short-term viability:** These cells can only survive for a limited time in the lab, so they are used for specific experiments rather than long-term studies.

- **Examples and Uses:**

- **Primary Monkey Kidney Cells:** These are often used to grow a variety of viruses, including **influenza**, **parainfluenza**, **enteroviruses**, and **adenoviruses**.
- **Human cells** like primary human fibroblasts may also be used for certain viral cultures.

Established (Finite) Cell Lines

- **What are Established Cell Lines?**

- Established or finite cell lines are **diploid** cells, just like primary cultures, but they can be kept growing in the lab for a longer time by carefully maintaining them.
- They are derived from tissues, but unlike primary cultures, they are **more stable** and can be cultured for several weeks or months.

- **How Are They Made?**

- Cells are cultured and selected for their ability to grow and divide in a controlled lab environment.
- Over time, these cells are adapted to grow under lab conditions but still retain the normal number of chromosomes.

- **Why Are They Important?**

- They are more **stable** than primary cultures, which makes them easier to work with for experiments that require longer periods.
- They are often used when **long-term virus studies** are needed but aren't as ideal for all types of viruses, as some viruses may not replicate well in them.

- **Examples and Uses:**

- **WI-38** (human lung fibroblast cells), **MRC-5** (human fetal lung cells), and **IMR-90** (human lung fibroblasts) are well-known established cell lines.
- These cell lines are used for studying viruses like **Hepatitis B**, **Herpes Simplex Virus**, and **Human Cytomegalovirus**.

Continuous Cell Lines

- **What are Continuous Cell Lines?**

- Continuous cell lines are cells that have been **transformed** so that they can grow and divide indefinitely. Unlike primary or established cells, these cells have an **abnormal number of chromosomes** and can survive for long periods without losing their ability to divide.
- These cells are often **heteroploid**, meaning they don't have the normal set of chromosomes found in the original tissue.

- **How Are They Made?**

- These cells are typically **transformed** (either naturally or artificially) by a virus or chemicals that cause them to grow indefinitely.
- They may be derived from tumors or cancerous tissues, which is why they can keep growing.

- **Why Are They Important?**

- Continuous cell lines are used for research that requires **long-term viral studies** because they can be maintained in the lab for an extended period.
- They are especially useful for studying viruses that require **high cell turnover** or when researchers need a constant supply of cells.

- **Examples and Uses:**

- **HeLa cells** (derived from human cervical cancer), **HEp-2 cells** (human epithelial cells), **A549 cells** (human lung carcinoma cells), and **Vero cells** (monkey kidney cells) are common continuous cell lines.
- These cells are ideal for growing viruses like **Herpes Simplex Virus (HSV)**, **Varicella Zoster Virus (VZV)**, **Cytomegalovirus (CMV)**, **Adenovirus**, and **Rhinoviruses** (common cold).

Cytopathic Effect (CPE)

- **What is a Cytopathic Effect (CPE)?**

- A Cytopathic Effect (CPE) refers to the changes or damage in cells that are infected by viruses. These changes can be **observed under a microscope** and help scientists identify which virus is causing the infection.
- CPE is not seen with all viruses. Some viruses replicate without causing visible damage to the infected cells.

- **What Are Some Common CPEs?**

- **Cell Clumping:** Infected cells may stick together or clump, which makes it easy to see under the microscope.
- **Vacuoles:** Empty, bubble-like spaces may appear inside the cells as they begin to break down.
- **Inclusions & Granules:** Small structures or particles may form inside the cell, indicating viral replication or interference with normal cell function.
- **Syncytium (Cell Fusion):** Multiple cells may fuse to form one large, multinucleated cell, which is a hallmark of certain viral infections like **measles** or **respiratory syncytial virus (RSV)**.
- **Cell Destruction:** Infected cells may burst or disintegrate, releasing new viral particles into the surrounding culture medium.

- **Why Is CPE Important?**

- By observing these changes, scientists can **identify which virus** is causing the infection and assess how the virus is affecting the cells.
- CPE helps **track viral growth** and determine the **severity** of viral infections in cell cultures.

Embryonated Eggs for Virus Growth

- **What Are Embryonated Eggs?**

- Embryonated eggs are **fertilized chicken eggs** that are used in laboratory settings to grow and study viruses.
- These eggs contain a developing **embryo**, which provides an ideal environment for the virus to replicate, especially for certain types of viruses that need a living organism to grow.

- **How Are They Used?**

- The eggs are typically **injected** with the virus under the egg's shell membrane or into specific parts of the egg, like the **allantoic cavity** or **amniotic fluid**, depending on the virus being studied.
- The virus then grows within the embryo as it develops, and researchers monitor the growth or effects of the virus.

- **Why Are Embryonated Eggs Important?**

- **Virus Growth:** Some viruses, particularly those that affect **animals or birds**, grow very well in embryonated eggs.
- The developing embryo's cells act as a host for viral replication.
- **Research & Vaccine Production:** Embryonated eggs are primarily used for research, **vaccine production**, and **virus cultivation**.
- For example, influenza vaccines are often produced using embryonated eggs.
- **Cost-Effective & Practical:** Compared to cell cultures, embryonated eggs are often more cost-effective and easier to maintain, particularly for large-scale virus production.

Embryonated Eggs for Virus Growth Conti...

- **Common Viruses Grown in Embryonated Eggs:**
 - **Influenza Virus:** One of the most common viruses cultivated in embryonated eggs, especially for the production of flu vaccines.
 - **Newcastle Disease Virus (NDV):** A virus affecting poultry, often studied in embryonated eggs.
 - **Avian Influenza Virus:** This virus, which affects birds, is commonly grown in eggs for both research and vaccine development.
 - **Yellow Fever Virus:** Embryonated eggs have been used for growing yellow fever virus to produce vaccines.
- **Why Not Use Eggs for Diagnostic Purposes?**
 - While eggs are extremely useful for growing viruses in the laboratory, they are **not typically used for diagnosing individual viral infections** in patients.
 - They are more commonly used for **large-scale virus production**, especially for vaccines, rather than for immediate diagnosis of infections.

Embryonated Eggs for Virus Growth Conti...

- **Advantages of Using Embryonated Eggs:**
 - **Efficient for Virus Growth:** Certain viruses, like the influenza virus, grow better in eggs than in other types of culture systems.
 - **Vaccine Production:** Embryonated eggs have been crucial for the mass production of vaccines, particularly for flu vaccines, making them an essential tool in **global health**.
- **Limitations of Using Embryonated Eggs:**
 - **Limited Virus Types:** Not all viruses can grow in embryonated eggs. Some viruses require cell cultures or animal models to propagate.
 - **Animal Welfare Concerns:** The use of animals, including fertilized eggs, raises ethical concerns, especially when alternative methods (like cell cultures) are available.

Animal Models

- **What are Animal Models?**

- Sometimes, scientists use live animals to study viruses.
- These animals can include mice, monkeys, or other species.

- **Why are they important?**

- Animal models are used in research to study how viruses infect living organisms and to test potential treatments or vaccines.
- They help researchers understand how a virus behaves in a whole body rather than just in cells or in a lab setting.

Electron Microscopy and Virus Detection:

- **Electron Microscopy (EM):**

- **Virion Size:** Most individual virions (virus particles) are too small to be observed with a regular light microscope.
- Electron microscopy is often required to see them, especially due to their tiny size.
- **Poxviruses:** These viruses are relatively large and are approximately the size of some small bacteria. This makes them visible under an electron microscope, whereas most viruses are too small for standard light microscopes.

- **Limitations of Electron Microscopy:**

- **Cost:** Electron microscopy is a costly technique, making it less accessible.
- **Expertise Required:** The technique demands high levels of expertise to correctly interpret the images.
- **Sensitivity:** It is generally not as sensitive as other modern viral detection methods.
- **Low Common Usage:** Because of its expense, technical demands, and low sensitivity, electron microscopy is not commonly used for routine virus detection, though it can still be helpful in specific cases.

General Characteristics of Herpesviruses

- **Shape & Structure:**

- Herpesviruses have an **icosahedral shape**, characterized by 20 triangular faces. This structural feature is common in many DNA viruses.
- They are **enveloped** viruses, meaning they have a lipid membrane derived from the host cell during viral replication.

- **Size:**

- Herpesviruses range in size from **90 to 100 nm**, which places them on the larger side among viruses, allowing them to carry a large genome.

- **Family:**

- These viruses belong to the family **Herpesviridae**, which includes multiple important human pathogens like **Herpes Simplex Virus (HSV)**, **Varicella-Zoster Virus (VZV)**, **Epstein-Barr Virus (EBV)**, and **Cytomegalovirus (CMV)**.

TABLE 9-1 FAMILIES OF DNA VIRUSES

Family	Important Human Viruses
<i>Poxviridae</i>	Variola virus, molluscum contagiosum virus
<i>Herpesviridae</i>	Herpes simplex viruses types 1 and 2, varicella-zoster virus, Epstein-Barr virus, cytomegalovirus, human herpesviruses 6, 7, and 8
<i>Adenoviridae</i>	Adenovirus
<i>Hepadnaviridae</i>	Hepatitis B virus
<i>Papillomaviridae</i>	Papillomavirus
<i>Polyomaviridae</i>	JC and BK viruses
<i>Parvoviridae</i>	Parvovirus B19

Transmission of Herpesviruses

- **Modes of Transmission:**

- **Direct Contact:** Herpesviruses are commonly transmitted through direct skin-to-skin contact, such as kissing, sexual contact, or contact with contaminated saliva.
- **Aerosol Droplets:** Viruses like **VZV** (Varicella) and **EBV** (mononucleosis) can also spread through respiratory droplets.
- **Vertical Transmission:** Some herpesviruses, like **CMV** and **HSV**, can be transmitted from mother to fetus during childbirth or in utero, leading to congenital infections.
- **Fomites:** Herpesviruses can persist on contaminated surfaces, potentially spreading through indirect contact.

- **Incubation Period:**

- The incubation period varies, with primary infections generally developing within **2 to 12 days** after exposure, depending on the virus and the individual's immune response.

Pathogenesis of Herpesviruses

- **Initial Infection & Replication:**

- Upon entry into the host via mucosal surfaces or breaks in the skin, herpesviruses infect epithelial cells, where they begin to replicate and cause primary lesions, such as cold sores in HSV or chickenpox in VZV.

- **Latent Infection:**

- After the initial infection, the virus migrates along sensory nerves and establishes latency in nerve ganglia (e.g., **trigeminal ganglion** for HSV, **dorsal root ganglion** for VZV).
- The virus remains dormant for years, evading the immune system by being quiescent in nerve cells.

- **Reactivation:**

- Stress, immunosuppression, or trauma can trigger reactivation of latent herpesviruses.
- Upon reactivation, the virus travels along the nerve axons back to the skin or mucosa, causing recurrent disease (e.g., cold sores in HSV, shingles in VZV).

- **Immune Evasion:**

- Herpesviruses have evolved numerous mechanisms to evade the host immune system, including inhibiting antigen presentation and downregulating immune responses.

Clinical Manifestations

- **Herpes Simplex Virus (HSV):**
 - **Primary Infection:** Can cause painful **oral or genital lesions**.
 - **HSV-1** typically causes oral infections (cold sores), while **HSV-2** is associated with genital herpes.
 - **Recurrent Infection:** Reactivation leads to **milder symptoms** localized to the site of latency.
 - **Severe Complications:** In immunocompromised patients, HSV can lead to **encephalitis, herpes keratitis** (eye infection), and disseminated disease.
- **Varicella-Zoster Virus (VZV):**
 - **Primary Infection (Chickenpox):** Characterized by a **vesicular rash** and systemic symptoms like fever.
 - **Reactivation (Shingles):** Results in a **painful, localized rash** typically in a dermatomal distribution. The pain (postherpetic neuralgia) can persist long after the rash resolves.

- **Epstein-Barr Virus (EBV):**

- Causes **infectious mononucleosis**, with symptoms including fever, sore throat, lymphadenopathy, and hepatosplenomegaly.
- EBV is also associated with cancers such as **Burkitt's lymphoma** and **Nasopharyngeal carcinoma**.

- **Cytomegalovirus (CMV):**

- Can cause **congenital infections**, leading to **microcephaly**, **hearing loss**, and developmental delay.
- In immunocompromised individuals (e.g., transplant recipients, HIV patients), CMV can cause severe disease like **pneumonitis** or **retinitis**.

Diagnosis of Herpesvirus Infections

- **Clinical Diagnosis:**

- Diagnosis is often based on the characteristic **clinical presentation**, such as the appearance of a rash or lesions in the case of VZV or HSV.

- **Laboratory Diagnosis:**

- **Polymerase Chain Reaction (PCR):** Highly sensitive for detecting viral DNA in lesions, blood, or cerebrospinal fluid.
- **Direct Fluorescent Antibody (DFA):** Used to detect viral antigens in lesions.
- **Serology:** Detection of **IgM** (indicating recent infection) or **IgG** (indicating past exposure) antibodies can help in diagnosing primary infections or reactivation.

Treatment & Management

- **Antiviral Therapy:**
 - **Acyclovir, Valacyclovir, and Famciclovir** are commonly used to treat herpesvirus infections. These drugs inhibit viral DNA synthesis.
 - In severe cases, intravenous acyclovir may be required.
- **Symptom Management for Shingles:**
 - **Analgesics and corticosteroids** are used to manage pain and inflammation associated with **shingles**.
 - Early antiviral therapy can reduce the severity of the infection.
- **Vaccines:**
 - Vaccines are available for **VZV** (Varicella and Shingles vaccines) and can help prevent primary infections and reduce the severity of reactivation in older adults.

Prevention and Public Health Considerations

- **Primary Prevention:**

- **Vaccination** against **VZV** (chickenpox) and **Shingles** can significantly reduce the incidence of primary infections and reactivations.
- **Safe sexual practices** (e.g., using condoms) and avoiding direct contact with active lesions can reduce the spread of **HSV** and **VZV**.

- **Control of Outbreaks:**

- In healthcare settings, strict isolation protocols may be needed for immunocompromised patients to prevent **CMV** or **HSV** infections.
- **CMV** infection can be a major concern in organ transplantation, necessitating antiviral prophylaxis and screening.

Summary Table

TABLE 9-2 MAJOR CLINICAL SYNDROMES OF HUMAN HERPESVIRUSES		
Virus	Major Clinical Syndrome	Site of Latent Infection
Herpes simplex virus Type 1	Gingivostomatitis in children and young adults, recurrent oral-labial infection (cold sores), infection of the cornea (keratitis), herpes encephalitis	Trigeminal nerve root ganglion and autonomic ganglia of superior cervical and vagus nerves
Type 2	Genital herpes, neonatal herpes	Sacral nerve root ganglia
Varicella-zoster	Chickenpox (primary infection), shingles or zoster (reactivation)	Thoracic, cervical or lumbar nerve root ganglia
Cytomegalovirus	Asymptomatic infection, heterophile-negative mononucleosis, fever hepatitis syndrome in neonates and transplant patients, interstitial pneumonia in immunocompromised patients	Leukocytes (neutrophils and lymphocytes)
Epstein-Barr virus	Heterophile-positive mononucleosis	B lymphocytes
Human herpesvirus 6	Roseola (sixth disease)	Peripheral blood mononuclear cells
Human herpesvirus 7	Roseola and febrile disease in children	Peripheral blood mononuclear cells
Human herpesvirus 8	Kaposi's sarcoma	Peripheral blood mononuclear cells

Summary

- Herpesviruses are **enveloped, icosahedral** DNA viruses that can cause both **acute** and **latent infections**.
- They are transmitted primarily through **direct contact** and **respiratory droplets**, with vertical transmission during birth.
- Herpesviruses exhibit a **latent phase** in nerve cells and can be reactivated under conditions of **stress** or **immunosuppression**.
- Management involves **antiviral therapy** and **vaccination**, with a focus on reducing symptoms and preventing complications like **shingles** or **CMV** in immunocompromised individuals.

Human Papillomavirus (HPV)

– General Characteristics

- **Shape & Structure:**
 - **Icosahedral shaped** virion.
 - **Enveloped** with a size range of **40 to 55 nm**.
- **Family:**
 - Belongs to the **Papillomaviridae** family.

HPV and Associated Diseases

- **Common Warts:**
 - **Plantar warts**, **genital warts**, and **flat warts** are caused by HPV infection.
- **Cervical Cancer:**
 - Certain **high-risk HPV strains** are associated with the development of **cervical cancer**.
 - HPV is the **most common cause** of cervical cancer worldwide.
- **Genital Warts:**
 - **Genital warts** are the **most common sexually transmitted disease (STD)** in the U.S.
 - HPV is the leading cause of **genital warts**.

Transmission of HPV

- **Transmission Routes:**

- Primarily transmitted through **sexual contact**, including vaginal, anal, and oral sex.
- Genital warts can also be spread through **skin-to-skin contact** in the genital area.

- **Incubation Period:**

- HPV can remain **asymptomatic** for years after infection, with some people showing no symptoms.

HPV Vaccination

- **HPV Vaccine:**

- **Recently approved vaccine** for females aimed at preventing **cervical cancer**.
- The vaccine contains the **strains of HPV** most commonly associated with **cervical cancer** (types 16, 18) and genital warts (types 6, 11).

- **Vaccination Impact:**

- The vaccine has shown to significantly reduce the incidence of **HPV-related cancers** and genital warts.

Poxviruses

– General Characteristics

- **Size and Shape:**

- Poxviruses are **large** viruses, ranging from **220 to 450 nm** in size.

- **Replication:**

- Poxviruses replicate entirely in the **cytoplasm** of the host cell.
 - Contain **DNA polymerase** for DNA replication and an **RNA polymerase** system for transcription.

- **Family:**

- Poxviruses belong to the **Poxviridae** family.

Variola Virus – Smallpox

- **Variola Virus (Smallpox):**
 - **Variola major** caused severe smallpox with a fatality rate of about **30%**.
 - **Variola minor** caused milder smallpox with a fatality rate of less than **1%**.
- **Smallpox Eradication:**
 - **World Health Organization** declared the world **smallpox-free in 1979** after a global vaccination program.
 - **Vaccinia virus** is used in the **smallpox vaccine**, which prevents **variola** infection.
- **Bioterrorism Risk:**
 - **Variola virus** is considered a potential **bioterrorism agent** due to its high fatality rate and potential for human-to-human transmission.

Other Poxviruses

- **Molluscipoxvirus:**

- Causes **molluscum contagiosum**, a **skin infection** that occurs worldwide.
- The infection is generally **self-limiting**, but it can cause **cosmetic concerns**.

- **Monkeypox Virus:**

- **Zoonotic virus** found primarily in **Africa**, transmitted from animals (mainly rodents) to humans.
- **Monkeypox** has caused outbreaks outside Africa, including a **recent outbreak in the U.S.**, traced back to **rodents** imported from Africa that transmitted the virus to prairie dogs.

Clinical Manifestations of Poxviruses

- **Smallpox:**
 - **High fever, rash, and vesicular lesions** that progress into scabs.
 - Potential for **severe complications** such as **hemorrhagic smallpox**, leading to death.
- **Molluscum Contagiosum:**
 - Causes **raised, pearl-like** bumps with a central dimple, usually on the face, arms, or genitals.
 - Common in children and can spread in **immunocompromised** individuals.
- **Monkeypox:**
 - Symptoms include **fever, rash, and swollen lymph nodes**.
 - The rash often begins on the **face** and then spreads to other parts of the body.

Prevention and Treatment of Poxviruses

- **Smallpox:**

- **Vaccination** with **vaccinia virus** was the key strategy in eradicating smallpox.
- **No treatment** for smallpox once infected; **supportive care** is provided to manage symptoms.

- **Molluscum Contagiosum:**

- Generally resolves on its own, but **topical treatments** (e.g., cryotherapy or curettage) can be used for severe cases.

- **Monkeypox:**

- **Supportive care** is the main treatment, and the **smallpox vaccine** provides some protection against monkeypox.

Summary

- **HPV:**

- Causes **genital warts**, **plantar warts**, and is associated with **cervical cancer**.
- A vaccine has been developed to prevent the most common strains associated with cancer.

- **Poxviruses:**

- Includes **Variola** (smallpox), **Molluscipoxvirus** (molluscum contagiosum), and **Monkeypox virus**.
- Smallpox has been eradicated, and vaccines prevent both smallpox and monkeypox.

Adenoviruses - Identifying Characteristics of Adenoviruses

- **Adenoviruses Overview:**
 - Adenoviruses are **naked (non-enveloped)**, meaning they do not have a lipid bilayer membrane, which makes them more resistant to environmental factors like drying and disinfectants.
 - **Shape:** These viruses are **icosahedral**, a geometric shape with 20 faces, giving them a symmetrical and compact structure.
 - **Size:** The diameter of adenoviruses ranges from **70 to 90 nm**, making them relatively small compared to other viruses.
- **Family:** Adenoviruses belong to the **Adenoviridae** family, which includes over 50 different types that infect humans and various animals.

Common Infections Caused by Adenoviruses

- **Respiratory Tract Infections:**

- Adenoviruses are a common cause of **upper respiratory infections** in **young children**.
- These infections can range from mild cold-like symptoms to more severe conditions like pneumonia and bronchiolitis.
- **Symptoms:** Fever, sore throat, runny nose, cough, and sometimes, pneumonia.

- **Urinary Tract and Gastrointestinal Infections:**

- Adenoviruses can infect the **urinary tract**, leading to conditions like **hemorrhagic cystitis** (painful, bloody urination).
- The virus can also cause **gastrointestinal issues**, including diarrhea and vomiting, particularly in children.

- **Pharyngitis:**

- In addition to respiratory infections, adenoviruses are often associated with **pharyngitis**, an inflammation of the throat that leads to a sore throat, fever, and difficulty swallowing.

Eye Infections and Keratoconjunctivitis

- **Eye Infections:**

- Adenoviruses can cause **eye infections**, which are more commonly seen in **newborns, immunocompromised individuals, and military recruits**.
- **Why Military Recruits?:** The virus spreads easily in environments with close living quarters, such as barracks, where outbreaks of eye infections are common.

- **Keratoconjunctivitis:**

- One of the most notable eye conditions caused by adenoviruses is **keratoconjunctivitis**, a serious infection that affects the cornea and conjunctiva of the eye. This condition can cause redness, swelling, pain, and, in severe cases, vision problems.
- Adenoviruses have been linked to **epidemics of keratoconjunctivitis**, particularly in closed environments like military camps or healthcare settings.

Sample Collection for Diagnosis

- **How are Adenoviruses Diagnosed?:**
 - **Throat Swabs:** Throat swabs are commonly used to collect samples from patients showing respiratory symptoms.
 - The sample is then tested for the presence of the virus.
 - **Eye Samples:** In cases of eye infections, samples can be taken from the eye (conjunctival swabs) to identify the virus.
 - **Stool Samples:** If gastrointestinal symptoms are present, **stool samples** may be tested for adenoviruses to confirm a gastrointestinal infection.

Medically Important RNA Viruses

Retroviruses

Structure:

- Icosahedral shape, enveloped virion.
- Size: 80 to 130 nm.
- **Key Feature:**
 - Contain **RNA-dependent DNA polymerase (reverse transcriptase)** for replication.
 - Reverse transcriptase uses viral RNA as a template to create **double-stranded DNA** which integrates into the host chromosome (called **provirus**).

Retroviruses - Replication Process

- **Provirus Formation:**
 - After reverse transcription, the newly formed double-stranded DNA moves into the host nucleus, integrating into the host genome.
 - The host's RNA polymerase transcribes the proviral genome into mRNA.

A. Lentiviruses - HIV

- **Human Immunodeficiency Virus (HIV):**
 - **HIV-1:** More prevalent and causes a more severe infection.
 - **HIV-2:** Less common but still important.
- **Spread:**
 - Sexual contact, intravenous drug use, mother-to-child transmission, and contaminated blood products.

1. HIV Pathogenesis

- **Initial Infection:**
 - The virus infects **macrophages** and **dendritic cells**, then **CD4+ T cells**, which are crucial for immune function.
 - **CD4** is the main receptor, with **CXCR4** and **CCR5** as important coreceptors.
- **Acute Infection:**
 - Symptoms are often mild and resemble **infectious mononucleosis**.
 - **Clinical latency** phase occurs: High virus replication but asymptomatic, with T cells being replenished.

2. HIV Progression

- **AIDS-Related Complex (ARC):**
 - T cell destruction exceeds replacement, leading to **chronic and recurrent infections** (e.g., Pneumocystis pneumonia, candidiasis, CMV infections).
- **AIDS (Acquired Immunodeficiency Syndrome):**
 - Severe immune compromise leads to life-threatening opportunistic infections and conditions like **Kaposi sarcoma** and **B cell lymphomas**.

3. HIV Diagnosis and Staging

- **Diagnosis:**
 - Clinical history, serology (ELISA), detection of viral antigens, or RNA.
- **Testing:**
 - **Rapid strip test** and **ELISA** for initial screening, **Western blot** for confirmation.
 - Staging based on **CD4+ cell count** and **opportunistic infections**.
 - **Viral load** can be determined via **RT-PCR**.

B. Human T-cell Lymphotropic Virus (HTLV)

- **HTLV Types:**

- **HTLV-1:** Linked to **adult T cell leukemia** and **HTLV-1-associated myelopathy (HAM/TSP)**.
- **HTLV-2:** No malignancy association, but linked to a neurological disease resembling HAM/TSP.

- **Transmission:**

- Sexual contact, breastfeeding, and intravenous drug use.

Orthomyxoviruses - Overview

Structure:

- **Segmented RNA genome, helical-shaped virion, enveloped.**
- Size: 75 to 125 nm.

Members:

- Influenza A, B, and C are the only members of the **Orthomyxoviridae** family.

Orthomyxovirus - Surface Proteins

- **Hemagglutinin (HA):**
 - Allows the virus to **attach to respiratory epithelial cells.**
 - Agglutinates red blood cells.
- **Neuraminidase (NA):**
 - Enzymatic activity helps release newly formed viral particles from infected cells.

Orthomyxovirus - Antigenic Drift & Shift

- **Antigenic Drift:**

- Point mutations in **HA** and **NA** genes.
- Causes gradual changes in the virus and may affect immunity.

- **Antigenic Shift:**

- Major change through **reassortment** of the RNA genome when a host cell is infected by multiple influenza strains.
- Only occurs in **influenza A**, can lead to **epidemics** and **pandemics**.

- **Influenza Vaccine**

- **Trivalent Influenza Vaccine:**

- The vaccine is updated yearly based on predicted prevalent strains for the upcoming season.

- **Orthomyxovirus Diagnosis**

- **Diagnosis:**

- Clinical symptoms, serology, **direct antigen detection**, and **viral isolation**.

Paramyxoviruses - Overview

- **Structure:**
 - **Helical-shaped** enveloped virions.
 - Size: 150 to 300 nm.
- **Families: Paramyxoviridae**, includes **morbilliviruses**, **pneumoviruses**, and **megamyxoviruses**.

1. Parainfluenza Viruses

- **Cause of Childhood Croup:**
 - Parainfluenza viruses are responsible for **croup**, a respiratory infection characterized by **fever** and a **hoarse cough**.

2. Mumps Virus

- **Mumps:**

- Mumps causes **swelling** of the **parotid glands**, leading to difficulty swallowing.
- Rare in developed countries due to the **vaccine**.

3. Measles (Rubeola)

- **Morbillivirus:**

- Causes **measles**, marked by **Koplik spots** (white lesions with a red border) on the **oral mucosa**.
- Nearly eliminated in developed countries due to vaccination programs.

4. Respiratory Syncytial Virus (RSV)

- **RSV:**

- Most common cause of **bronchitis** and **pneumonia** in infants and young children.
- **Syncytia formation:** Infected cells fuse with adjacent cells, forming multinucleated giant cells.
- **RSV Diagnosis and Treatment**
 - **Diagnosis:**
 - Can be grown in **human heteroploid cells** like HEp-2, HeLa, and A549.
 - Diagnosis is primarily based on **clinical symptoms** and lab tests.

Picornaviruses Overview

- **Characteristics:**

- Small, **naked** virions (20 to 30 nm).
- Family **Picornaviridae** includes several important viruses: **Enteroviruses**, **Hepatitis A Virus (HAV)**, and **Rhinovirus**

- **Characteristics:**

- Small, **naked** virions (20 to 30 nm).
- Family **Picornaviridae** includes several important viruses: **Enteroviruses**, **Hepatitis A Virus (HAV)**, and **Rhinoviruses**.

1: Enteroviruses

- **Common Infections:**

- Cause a variety of infections, including **fever**, **respiratory**, and **gastrointestinal** infections.
- Examples: **Poliovirus**, **Coxsackieviruses**, and **Echoviruses**.

- **Diagnosis:**

- **Nucleic acid amplification** tests (e.g., **serum**, **CSF**, **throat swabs**, **rectal swabs**).

2: Poliovirus

- **Transmission:**

- Spread via **fecal-oral route**.
- Infects the **gastrointestinal tract** and can spread to the **CNS**.

- **Clinical Manifestations:**

- Most cases are **mild**, but can lead to **meningitis** or **paralytic polio**.

- **Vaccines:**

- **Salk vaccine:** Inactivated (formalin-treated).
- **Sabin vaccine:** Live-attenuated.
- Due to successful vaccination programs, **polio risk is very low** globally.

3: Coxsackievirus

- **Coxsackie A Viruses:**

- Cause **hand, foot, and mouth disease** (distinct from **foot and mouth disease** in animals).
- Associated with **conjunctivitis**.

- **Coxsackie B Viruses:**

- Cause **myocarditis** (1/3 of cases).
- Associated with **meningitis**.

4: Rhinoviruses

- **Cause:**

- The most frequent cause of the **common cold**.
- **100+ serotypes** - No cross-protection between serotypes, leading to frequent infections.

- **Growth Conditions:**

- Thrive at **33°C**, which is near the temperature of the **nasal passage**.

- **Prevention:**

- **Handwashing** and avoiding **hand-to-nose** contact.

5: Rotaviruses Overview

- **Characteristics:**
 - **Double-stranded RNA** genome.
 - Virion size: **70 nm**, with a **wheel-like appearance**.
- **Important Infections:**
 - **Most significant cause of gastrointestinal infections** in children under **2 years**.
- **Diagnosis:**
 - **Antigen detection** (via **latex agglutination** or **ELISA**).
 - Less common: **Immunoelectron microscopy**.

Caliciviruses Overview

- **Family: Caliciviridae** includes **Norovirus**, **Sapovirus**, **Lagovirus**, and **Vesivirus**.
- **Key Virus:**
 - **Norovirus** (also known as **Norwalk virus**), causes **highly contagious** gastroenteritis.

Togaviruses Overview

- **Characteristics:**
 - Virions are **70 nm, enveloped**.
- **Genera:**
 - **Rubivirus** (includes **Rubella virus**).
 - **Alphavirus** (contains ~25 viruses, all transmitted by **arthropods**).

1: Rubella (Rubivirus)

- **Disease: Rubella** (German measles).
 - **Mild** infection in children, but **severe congenital infections** in early pregnancy.
 - **Congenital Rubella Syndrome (CRS)** can cause **birth defects**.
- **Prevention:**
 - **Vaccination** programs have almost eliminated Rubella in developed countries.

2: Alphavirus

- **Transmission: Arthropod-borne** (e.g., mosquitoes).
- **Important Infections:** Includes viruses such as **Eastern Equine Encephalitis** and **Chikungunya virus**

Flaviviruses Overview

- **Family: Flaviviridae** (many are **arboviruses**).
- **Important Members:**
 - **West Nile virus, St. Louis encephalitis virus, Yellow fever virus, Dengue virus.**

1: West Nile Virus (WNV)

- **Transmission:**
 - **Birds** are the primary reservoir, transmitted to humans via **mosquitoes**.
- **Symptoms:**
 - Typically **mild or asymptomatic**, but **fatal encephalitis** is a severe complication.
- **Diagnosis:**
 - **ELISA antigen capture, RT-PCR.**

Rhabdoviruses Overview

- **Family: Rhabdoviridae.**
- **Characteristics:**
 - Bullet-shaped, **enveloped capsid.**
 - Size: **150-350 nm.**
- **Important Virus: Rabies virus (Genus Lyssavirus).**

1: Rabies (Rhabdoviridae)

- **Transmission:**
 - Through animal bites (e.g., **cats, dogs, raccoons**) or **bat contact.**
- **Pathogenesis:**
 - Virus infects **muscle tissue**, migrates to **neurons**, and spreads to the **CNS.**
 - **Progresses to fatal encephalitis.**
- **Diagnosis:**
 - **Direct fluorescent-antibody test.**
 - Detection of **Negri bodies** (virus inclusions in brain cells).

Rabies Vaccination

- **Prevention:**
 - **Rabies vaccine** is available for those at risk.
 - **Post-exposure treatment** is effective if administered within **72 hours**.
 - **Without treatment**, the infection is nearly always **fatal**.

Filoviruses Overview

- **Family:** Filoviridae.
- **Key Viruses:** **Marburg** and **Ebola** viruses.
- **Transmission:** Reservoirs are believed to be **bats**, with unclear transmission routes.
- **Clinical Outcome:**
 - Hemorrhagic fever, with **high fatality rates**.
- **Geographic Range:**
 - Primarily found in **Africa**.

Hepatitis A Virus (HAV)

- **Characteristics:**
 - Icosahedral, naked virion.
 - Size: **24-30 nm**.
 - Family: **Picornaviridae**, Genus: **Hepatovirus**.
- **Transmission:**
 - Spread via **fecal-oral route**.
 - Commonly caused by **contaminated water or food**.
- **Clinical Outcome:**
 - Acute liver infection, with **jaundice** and **flu-like symptoms**.
 - **No chronic infection**.

Hepatitis A Virus (HAV) Diagnosis

- **Diagnosis:**
 - Detection of **HAV-specific antibodies** (IgM and IgG) in blood.
 - **ELISA** and other antigen-detection methods.

TABLE 9-4 IMPORTANT HUMAN HEPATITIS VIRUSES

	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Family	<i>Picornaviridae</i>	<i>Hepadnaviridae</i>	<i>Flaviviridae</i>	Unclassified	<i>Hepeviridae</i>
Genome	RNA	DNA	RNA	RNA	RNA
Transmission	Fecal-oral	Parenteral, blood, sexually, needles, perinatal	Parenteral, blood, needles, perinatal	Parenteral, blood, sexually, needles, perinatal	Fecal-oral
Comments	No chronic liver disease, rarely fatal, severity increases with age	5–10% chronic hepatitis, associated with hepatocellular cancer	Chronic infections are common	Coinfection/ superinfection in patients infected with HBV	Wide range of clinical outcomes, high mortality rate in pregnant women

Hepatitis A Virus (HAV)

Hepatitis A Virus (HAV) Overview

- **Family:** Picornaviridae
- **Virion Size:** 24-30 nm, **Naked** (icosahedral shape)
- **Transmission:**
 - Fecal-oral route (poor sanitation, food handling, contaminated shellfish).
- **Vaccines:** Available for prevention.

HAV Clinical Characteristics

- **Incubation Period:** 15-40 days.
- **Symptoms:**
 - **Liver involvement** (jaundice).
 - **Nausea, anorexia, malaise.**
- **Mortality Rate:** Less than 1%.

HAV Diagnosis

- **Clinical Symptoms** and elevated **liver enzymes** (especially alanine aminotransferase).
- **Serologic Indicators:**
 - **Anti-HAV IgM:** Positive in acute infections.
 - **Anti-HAV IgG:** Indicates past infection or immunity.
 - **Ruling out** Hepatitis B virus in serology

Hepatitis B Virus (HBV)

- **Hepatitis B Virus (HBV) Overview**

- **Family:** Hepadnaviridae
- **Virion Size:** 42-47 nm (enveloped, **Dane particles**).
- **Genomic Structure:** Partially double-stranded DNA, requires an RNA intermediate for replication.
- **Transmission:** Blood, sexual transmission, contaminated needles, tattoos, intravenous drug use, etc.
- **Vaccines:** Available and recommended for healthcare workers.

- **HBV Clinical Characteristics**

- **Incubation Period:** 50-180 days.
- **Acute Infections:**
 - Symptoms resembling **HAV** infection.
- **Chronic Infections:**
 - Can lead to **cirrhosis** and **hepatocellular carcinoma**

- **HBV Diagnosis**

- **Clinical Symptoms** and elevated liver enzymes.
- **Serologic Markers:**
 - **HBsAg** (Hepatitis B surface antigen): First marker, indicates infectiousness.
 - **Anti-HBs:** Indicates recovery or immunity after vaccination.
 - **Anti-HBc IgM:** Recent acute infection (appears during "core window").
 - **Total Anti-HBc:** Indicates past or current infection.
 - **HBeAg:** Indicates high infectivity.
 - **Anti-HBe:** Good prognosis, indicates resolution of infection

Hepatitis D Virus (HDV) Overview

- **Family: Delta virus** (RNA virus).
- **Virion Size:** 35-37 nm (naked).
- **Key Points:**
 - Requires **HBsAg** to replicate.
 - **Coinfection:** Acquired simultaneously with **HBV**.
 - **Superinfection:** Occurs when an HBV-infected individual gets exposed to HDV (more severe).

• HDV Diagnosis

- **Detection of Anti-HDV and HDV RNA.**
- **Co-infection** and **Superinfection** markers will also be positive for **HBV** (e.g., **HBsAg**).

Hepatitis C Virus

Hepatitis C Virus (HCV) Overview

- **Family: Flaviviridae, Genus: Hepacivirus**
- **Virion Structure:** RNA virus, **lipid envelope**.
- **Transmission:**
 - Bloodborne (contaminated blood products, organ transplants, dialysis, IV drug use).
- **No Vaccine:** Currently no vaccine available for prevention.

• HCV Clinical Characteristics

- **Incubation Period:** 2-25 weeks.
- **Acute HCV Infection:**
 - Often **mild** or **asymptomatic**.
- **Chronic Infection:**
 - Commonly leads to **chronic hepatitis, cirrhosis**, and is one of the leading causes of **liver transplants** in the U.S.

• HCV Diagnosis

- **Elevated liver enzymes.**
- **Serologic Testing:**
 - **Anti-HCV** and **HCV antigen**.
 - **Nucleic acid amplification** for HCV RNA.
- **Cell culture:** The virus has not been grown in cell culture.

Other Human Hepatitis Viruses

Hepatitis E virus (HEV)

- HEV contains RNA, and virions range in size from 32 to 34 nm.
- HEV is spread by the fecal-oral route, often in contaminated water. It is the
- most common cause of hepatitis in some countries with poor sanitation.
- Diagnosis: Serology

Hepatitis G virus (HGV)

- HGV contains RNA and has an envelope.
- It is in the same family, Flaviviridae, as HCV.

HGV transmission

- contact with blood,
- sexually contact
- mother to child.
- Infection seems to be relatively common worldwide, but HGV is believed to be nonpathogenic.

Viral Drugs Function and Drug Resistance Mechanisms in Humans Viruses

Introduction

- **Drug resistance** is a significant challenge in treating viral infections.
- Resistance arises when a virus mutates and evades the effects of drugs meant to inhibit its replication.
- Key viruses: **HIV, Influenza, Hepatitis B, Hepatitis C, and Herpes Simplex Virus (HSV).**

Viral Life Cycle and Drug Targets

- **Overview of Viral Replication Stages:**
 - **Attachment and Entry:** Virus binds to and enters host cell.
 - **Replication:** Virus genome is copied.
 - **Protein Translation:** Viral proteins are made.
 - **Assembly and Budding:** New viral particles are assembled and released.
- **Antiviral drugs** target specific stages in this cycle:
 - a. **Entry Inhibitors:** Block virus entry into host cells (e.g., Maraviroc for HIV).
 - b. **Polymerase Inhibitors:** Block replication of viral genetic material (e.g., Acyclovir for HSV).
 - c. **Protease Inhibitors:** Prevent virus assembly by inhibiting viral protease (e.g., Lopinavir for HIV).
 - d. **Neuraminidase Inhibitors:** Block the release of new viral particles (e.g., Oseltamivir for Influenza).

General Concepts of Drug Resistance

- **Drug Resistance:** When a virus evolves to evade the effects of drugs.
- Resistance occurs due to **mutations** in viral genes, making the virus less susceptible or immune to drugs.
- **Viral Mutation:** High mutation rates in viruses, especially RNA viruses (like HIV and Influenza), contribute to the rapid development of resistance.

Mechanisms of Drug Resistance

Viruses can become resistant to drugs via various mechanisms

A. Mutation of Target Gene

- **Explanation:** The drug targets a specific viral protein or enzyme. A mutation in the gene encoding this protein can alter its structure, making the drug less effective or completely ineffective.
- **Example:** In **HIV**, mutations in the **reverse transcriptase** gene can reduce the effectiveness of reverse transcriptase inhibitors (RTIs).

B. Altered Viral Enzyme Activity

- **Explanation:** Some drugs target viral enzymes like **protease** or **polymerase**, but mutations can change the enzyme structure, reducing the drug's ability to inhibit its function.
- **Example:** **Influenza** virus can mutate **neuraminidase**, making it resistant to drugs like **oseltamivir**.

Mechanisms of Drug Resistance

C. Enhanced Drug Efflux

- **Explanation:** The virus may increase the activity of **efflux pumps**, which are proteins that expel antiviral drugs from the host cell before they can have an effect.
- **Example:** **HIV** and **Hepatitis B** can enhance efflux pumps to reduce drug concentrations inside infected cells.

D. Reduced Drug Entry

- **Explanation:** Viruses can alter the host cell's receptors or membrane proteins, making it harder for drugs to enter the cell and reach their targets.
- **Example:** In **HIV**, mutations in the **CD4 receptor** or coreceptors may reduce the ability of entry inhibitors to prevent viral attachment.

E. Viral Polymerase Mutations

- **Explanation:** Mutations in the viral polymerase gene can lead to changes that make the polymerase less sensitive to polymerase inhibitors, which prevent viral replication.
- **Example:** **Hepatitis C** virus may develop resistance to **NS5B polymerase inhibitors** due to mutations in the polymerase gene.

Drug Resistance in HIV

- **Resistance Mechanisms:**
 - **Reverse Transcriptase Inhibitors (RTIs):** Mutations in the reverse transcriptase gene reduce drug binding.
 - **Protease Inhibitors:** Mutations in the protease gene cause reduced drug effectiveness.
 - **Integrase Inhibitors:** Mutations in the integrase enzyme confer resistance to integrase strand transfer inhibitors (INSTIs).
- **Prevention:** Combination therapy (ART) combines multiple drugs, making it less likely that the virus will simultaneously develop resistance to all drugs.

Drug Resistance in Influenza

- **Resistance Mechanisms:**
 - **Neuraminidase Mutations:** Changes in the **neuraminidase** enzyme reduce the ability of drugs like **oseltamivir** to bind and inhibit the virus.
 - **Polymerase Mutations:** Mutations in the viral **polymerase** complex can lead to resistance to drugs like **baloxavir**.
- **Prevention:** Continuous monitoring of circulating strains and appropriate use of antiviral medications can help reduce resistance.

Drug Resistance in Hepatitis C

- **Resistance Mechanisms:**
 - **NS3/4A Protease Inhibitors:** Mutations in the protease gene lead to reduced susceptibility.
 - **NS5A Inhibitors:** Resistance due to mutations in the NS5A gene, which is targeted by drugs like **ledipasvir** and **daclatasvir**.
 - **NS5B Polymerase Inhibitors:** Mutations in the polymerase gene reduce drug effectiveness.
- **Treatment:** Direct-acting antivirals (DAAs) are the current standard, but resistance can develop if therapy is not completed properly.

Drug Resistance in Hepatitis B

- **Resistance Mechanisms:**
 - **Nucleos(t)ide Analogs:** Resistance arises due to mutations in the **polymerase** gene, reducing the drugs' ability to inhibit viral replication.
 - **Lamivudine and Entecavir Resistance:** Long-term use can result in resistance, especially in immunocompromised individuals.
- **Management:** Resistance testing and switching to alternative treatments are common approaches.

Drug Resistance in Herpes Simplex Virus (HSV)

- **Resistance Mechanisms:**
 - **Thymidine Kinase Mutations:** A mutation in the gene encoding **thymidine kinase** prevents the activation of acyclovir, a common treatment for HSV.
 - **DNA Polymerase Mutations:** Mutations in the DNA polymerase gene reduce susceptibility to acyclovir and other antiviral drugs.
- **Prevention:** Early detection and alternative antiviral agents can help manage resistance.

Factors Contributing to Drug Resistance

- **High Mutation Rates:** RNA viruses like HIV and Influenza mutate rapidly, increasing the likelihood of resistance.
- **Suboptimal Drug Dosing:** Irregular or inadequate dosing of antiviral drugs can allow resistant strains to survive and proliferate.
- **Co-infection:** Infections with multiple viruses (e.g., HIV + Hepatitis B or C) can promote resistance.
- **Immunocompromised Hosts:** Patients with weakened immune systems (e.g., HIV patients on ART) are more likely to develop resistance due to prolonged viral replication.

Clinical Implications and Management

- **Drug Resistance Testing:** Resistance testing is critical for choosing the right antiviral therapy and avoiding ineffective drugs.
- **Combination Therapy:** Using multiple drugs from different classes reduces the risk of resistance.
- **Monitoring Viral Load:** Regular monitoring helps assess treatment effectiveness and detect resistance early.
- **New Drug Development:** Ongoing research into second-line therapies and novel drug targets is essential.

Future Directions in Addressing Drug Resistance

- **Vaccine Development:** Vaccines like the flu vaccine reduce the need for antiviral drugs, preventing resistance.
- **Personalized Medicine:** Tailoring therapy based on an individual's resistance profile improves treatment outcomes.
- **Next-Generation Antivirals:** Developing drugs that target viral functions less likely to mutate (e.g., viral assembly, entry) reduces resistance.
- **Broad-Spectrum Antiviral Drugs:** New drugs that target multiple viruses can reduce resistance across different infections.